disastrous consequences of over-population. It is the basic assumption of this book that survival machines in general are guided by selfish genes, who most certainly cannot be expected to see into the future, nor to have the welfare of the whole species at heart. This is where Wynne-Edwards parts company with orthodox evolutionary theorists. He thinks there is a way in which genuine altruistic birth control can evolve.

A point that is not emphasized in the writings of Wynne-Edwards, or in Ardrey's popularization of his views, is that there is a large body of agreed facts that are not in dispute. It is an obvious fact that wild animal populations do not grow at the astronomical rates of which they are theoretically capable. Sometimes wild animal populations remain rather stable, with birth rates and death rates roughly keeping pace with each other. In many cases, lemmings being a famous example, the population fluctuates widely, with violent explosions alternating with crashes and near extinction. Occasionally the result is outright extinction, at least of the population in a local area. Sometimes, as in the case of the Canadian lynx—where estimates are obtained from the numbers of pelts sold by the Hudson's Bay Company in successive years—the population seems to oscillate rhythmically. The one thing animal populations do not do is go on increasing indefinitely.

Wild animals almost never die of old age: starvation, disease, or predators catch up with them long before they become really senile. Until recently this was true of man too. Most animals die in childhood, many never get beyond the egg stage. Starvation and other causes of death are the ultimate reasons why populations cannot increase indefinitely. But as we have seen for our own species, there is no necessary reason why it ever has to come to that. If only animals would regulate their *birth rates*, starvation need never happen. It is Wynne-Edwards's thesis that that is exactly what they do. But even here there is less disagreement than you might think from reading his book. Adherents of the selfish gene theory would readily agree that animals *do* regulate their birth rates. Any given species tends to have a rather fixed clutch size or litter size: no animal has an infinite number of children. The disagreement comes not over *whether* birth rates are regulated. The disagreement is over *why* they are regulated: by what process of natural selection has family planning evolved? In a nutshell, the disagreement is over whether animal birth control is altruistic, practised for the good of